Final Year Project Report

Full Unit - Final Report

A Smart IoT Solution for Monitoring and Tracking Small Sailing Boats in Mogadishu Coastal Areas.

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A report submitted in partial fulfilment of the degree of

Bachelor Of Science in Information technology

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DECLARATION: A

"We declare that the following is our own work and does not contain any unacknowledged work from any other sources. This project was undertaken to fulfill the requirements of the bachelor's degree program in Information Technology at SIMAD University."

Student One:

Mohamed Ali Ahmed

I would like to express my deepest gratitude to Almighty Allah for granting me the strength, knowledge, and perseverance to complete this project. My sincere thanks go to our supervisor, Mr. Abdulaziz Yasin Nageye, for his invaluable guidance, support, and encouragement throughout this research. I am also grateful to my family for their continuous support and prayers. Lastly, I appreciate my teammate for the cooperation and commitment that made this project possible.

Student Two:

Saabir Mohamed Mohamud

First and foremost, I thank Allah for His endless mercy and blessings throughout this journey. I would also like to extend my heartfelt appreciation to our supervisor Mr. Abdiaziz Yasin Nageeye whose insightful feedback and mentorship played a significant role in the success of this project. I am thankful to my family for their encouragement and support. My appreciation also goes to my project partner for the dedication and teamwork throughout the research

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DECLARATION. B

SUPERVISOR'S WORDS

I, on this day, declare that the final year project report entitled "A Smart IoT Solution for Monitoring and Tracking Small Sailing Boats in Mogadishu Coastal Areas" is a product of the student's originality and has been conducted independently under the supervision of Mr. Abdulaziiz Yasin Nageye , Lecturer and Research Supervisor at SIMAD University.

This report has not been presented elsewhere for the award of any degree, diploma, or certificate. All sources and literature utilized in this research have been appropriately acknowledged and referenced. I declare that the content of this project is in agreement with the research and development activities concerning the monitoring and tracking challenges of small sailing boats in Mogadishu using IoT Technology.

Supervisor : ___________________________

Signature : ____________________________

Date : ______________________________

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I thank Allah, the Creator of this world, for giving me the strength, patience, and opportunity to complete this project successfully. Success is never achieved alone; behind every achievement, there are people whose support, guidance, and encouragement make it possible.

I am deeply grateful to my family and parents for their unwavering emotional and financial support throughout this journey. Their belief in me has been my greatest motivation and the foundation of my success.

A special thanks to my supervisor, Mr.Abdulazizi Yasin Nageye , for his immense support, invaluable advice, and continuous guidance throughout this project. His insights and encouragement have been instrumental in shaping this work and bringing it to completion.

I would also like to express my appreciation to my friends and colleagues who have provided support, motivation, and constructive feedback. Their encouragement and collaboration have been truly invaluable.

Finally, I extend my gratitude to everyone who contributed their time, expertise, and resources to make this research possible. Your generosity and support have played a crucial role in the successful completion of this project.

ABSTRACT

Small sailing boats and fishing vessels are a vital component of the blue economy in Mogadishu, Somalia, providing livelihoods for many and supporting local trade. However, these boats face significant challenges, including the risk of theft, getting lost at sea due to unpredictable weather, and the lack of real-time monitoring systems in coastal waters. Traditional safety measures are often insufficient for tracking these vessels once they leave the shore. This project is designed to address these challenges by developing "A Smart IoT Solution for Monitoring and Tracking Small Sailing Boats in Mogadishu Coastal Areas."

The proposed system utilizes an Arduino Uno microcontroller integrated with a GPS (NEO-6M) module and a GSM (SIM800L) module to provide precise, real-time geographic coordinates of the boat. The system is designed to enhance maritime safety by allowing boat owners and port authorities to monitor the vessel's location via SMS alerts and instant communication. Key features include geofencing capabilities, which alert the owner if the boat drifts outside a designated safe zone, and an emergency SOS button that sends immediate distress signals with location data in case of engine failure or security threats.

Furthermore, the system incorporates sensors to detect unauthorized movement or tampering while the boat is docked. Field testing was conducted along the Mogadishu coastline to ensure signal reliability and tracking accuracy under maritime conditions. The results demonstrate that the IoT solution provides a cost-effective and reliable method for improving the security and operational monitoring of small sailing boats.

Keywords: Smart IoT Solution, Maritime Security, Sailing Boats, GPS Tracking, GSM Communication, Mogadishu Coastal Areas, Vessel Monitoring, Somalia.

LIST OF ABBREVIATIONS AND SYMBOLS

Abbreviation	Full Meaning
loT	Internet of Things
GPS	Global Positioning System
GSM	Global System for Mobile Communications
GPRS	General Packet Radio Service
SMS	Short Message Service
SIM	Subscriber Identity Module
UML	Unified Modeling Language
IDE	Integrated Development Environment (e.g., Arduino IDE
MCU	Microcontroller Unit
Li-PO	Lithium-ion Polymer (Battery)
SOS	Save Our Souls (Distress Signal)
LED	Light Emitting Diode
DC	Direct Current
UART	Universal Asynchronous Receiver-Transmitter
HTTP	Hypertext Transfer Protocol
API	Application Programming Interface

LIST OF SYMBOLS / UNITS

Symbol	Description / Unit
V	Voltage (Volt)
A	Ampere (Current)
mAh	Milliampere-hour (Battery Capacity)
Hz	Hertz (Frequency)
°	Degree (Used for Latitude/Longitude)
W	Watt (Power)
Ω	Ohm (Resistance)
Λ	Wavelength

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Chapter One

1.1 Introduction

Small sailing boats are central to Mogadishu's coastal economy, serving as the backbone of artisanal fishing and local transport. These vessels sustain livelihoods, provide food security, and support the local economy. However, the absence of modern monitoring systems exposes them to serious risks such as theft, accidents, and delayed rescue operations. Traditional methods like manual observation or radio communication are insufficient, leaving boat operators vulnerable in unpredictable weather and security-challenged environments (Landge et al., 2025).

Globally, the maritime industry has embraced **Internet of Things (IoT)** technologies to address similar challenges. IoT systems integrate GPS, sensors, and communication protocols to enable real-time tracking, environmental monitoring, and predictive analytics. For instance, **LoRaWAN-based IoT systems** have been deployed in Southeast Asia to monitor fishing boats, offering long-range communication at low cost (Neumann, 2025; Lyimo et al., 2025). Likewise, **ESP32 microcontrollers and solar-powered sensors** have been adopted in developing regions to reduce reliance on expensive satellite systems (Lin, 2026; Shwetha et al., 2026). These innovations demonstrate that IoT can be adapted to resource-constrained environments like Mogadishu.

In Somalia, maritime authorities face difficulties in regulating coastal activities due to the absence of centralized monitoring systems. This gap results in poor emergency response, inefficient resource management, and limited data for policy-making. The lack of reliable communication infrastructure further compounds risks, especially during storms or piracy incidents (Ministry of Fisheries Somalia, 2022). Implementing IoT-based solutions would not only improve safety but also enhance governance by providing accurate, real-time data to authorities.

Moreover, IoT applications extend beyond safety. They can support fishermen by providing information on weather conditions, sea currents, and potential fishing zones. Studies show that IoT-enabled dashboards can increase fishing productivity by up to 20% in regions where traditional methods dominate (Freire et al., 2022). For Mogadishu, this means greater economic resilience and improved food security.

In summary, the introduction highlights the urgent need for modern monitoring systems in Mogadishu's coastal areas. By leveraging IoT technologies such as GPS, ESP32 microcontrollers, and cloud dashboards, Somalia can bridge the gap between traditional practices and modern maritime safety standards. This research builds on global experiences while tailoring solutions to the unique challenges of Mogadishu's small sailing boats.

1.2 Background of the Study

Mogadishu's coastline stretches along the Indian Ocean and is home to hundreds of small sailing boats that serve as the backbone of artisanal fishing and local transport. These vessels are critical for sustaining livelihoods, providing food security, and supporting the local economy. However, they operate in an environment characterized by limited infrastructure, unpredictable weather, and security challenges. Traditional monitoring methods—such as manual observation or radio communication—are insufficient, leaving boat operators vulnerable to accidents, theft, and delayed rescue operations (Landge et al., 2025).

Globally, the maritime industry has embraced **IoT technologies** to address similar challenges. IoT systems integrate GPS, sensors, and communication protocols to enable real-time tracking, environmental monitoring, and predictive analytics. For example, **LoRaWAN-based IoT systems** have been deployed in Southeast Asia to monitor fishing boats, providing long-range communication at low cost (Neumann, 2025; Lyimo et al., 2025). Similarly, **ESP32 microcontrollers and solar-powered sensors** have been adopted in developing regions to reduce reliance on expensive satellite systems (Lin, 2026; Shwetha et al., 2026). These innovations demonstrate that IoT can be adapted to resource-constrained environments like Mogadishu.

In Somalia, maritime authorities face difficulties in regulating coastal activities due to the absence of centralized monitoring systems. This gap results in poor emergency response, inefficient resource management, and limited data for policy-making. The lack of reliable communication infrastructure further compounds the risks faced by small sailing boats, especially during storms or piracy incidents (Ministry of Fisheries Somalia, 2022). Implementing IoT-based solutions would not only improve safety but also enhance governance by providing accurate, real-time data to authorities.

Moreover, IoT applications extend beyond safety. They can support fishermen by providing information on weather conditions, sea currents, and potential fishing zones. Studies show that IoT-enabled dashboards can increase fishing productivity by up to 20% in regions where traditional methods dominate (Freire et al., 2022). For Mogadishu, this means greater economic resilience and improved food security.

The socio-economic context of Mogadishu makes IoT particularly suitable. Devices such as GPS trackers, ESP32 modules, and cloud dashboards are affordable, energy-efficient, and scalable. They can be deployed incrementally, starting with pilot projects in high-risk zones before expanding to cover the entire coastline. This phased approach ensures sustainability and community acceptance (Lin, 2026).

In summary, the background of this study highlights the urgent need for modern monitoring systems in Mogadishu's coastal areas. By leveraging IoT technologies, Somalia can bridge the gap between traditional practices and modern maritime safety standards. This research builds on global experiences while tailoring solutions to the unique challenges of Mogadishu's small sailing boats.

1.3 Problem Statement

Small sailing boats in Mogadishu's coastal areas are essential for fishing and local transport, yet they face serious risks due to the absence of modern monitoring systems. Currently, there is no reliable real-time tracking of these vessels, which means that when accidents occur, rescue operations are delayed and often ineffective. The lack of communication equipment also exposes boat operators to dangers such as storms, piracy, and mechanical failures, with little chance of timely assistance (Landge et al., 2025).

In addition, maritime authorities struggle to regulate coastal activities because of limited data availability. Without accurate information on boat movements, it becomes difficult to enforce safety standards, manage resources, or respond to emergencies. This gap in monitoring contributes to frequent loss of lives, property damage, and reduced economic productivity in the fishing sector (Ministry of Fisheries Somalia, 2022).

Therefore, the problem is the absence of a centralized, automated IoT-based monitoring system for small sailing boats in Mogadishu. Addressing this issue is critical to improving safety, supporting fishermen, and strengthening maritime governance (Neumann, 2025).

1.4 Research Objectives

1.4.1 General Objective

The general objective of this study is to design and develop an IoT-based monitoring system for small sailing boats in Mogadishu's coastal waters.

1.4.2 Specific Objectives

The study aims to:

1. Integrate GPS and sensor networks for real-time tracking and environmental monitoring.
2. Evaluate system performance in Mogadishu's coastal waters.
3. Enhance safety and efficiency of small-scale maritime operations through IoT-based solutions (Lyimo et al., 2025).

1.5 Scope of the Study

This study is geographically limited to the coastal areas of Mogadishu, where small sailing boats are widely used for fishing and short-distance transport. The focus is specifically on artisanal and community-level vessels, excluding large commercial ships, naval fleets, or international cargo vessels. By narrowing the scope to small boats, the research directly addresses the most vulnerable group in Somalia's maritime sector, which often lacks access to advanced navigation and safety technologies (Landge et al., 2025).

Technologically, the scope of this study emphasizes the design and testing of a low-cost IoT prototype. The system will integrate GPS modules for location tracking, ESP32 microcontrollers for processing, and LoRaWAN or GSM/GPRS protocols for communication. A cloud-based dashboard will be developed to visualize real-time boat movements and environmental data. The study does not extend to advanced predictive analytics or large-scale maritime logistics but instead focuses on practical, affordable solutions that can be implemented in Mogadishu's socio-economic context (Lin, 2026; Shwetha et al., 2026).

The scope also includes evaluating the performance and feasibility of the IoT solution in real-world conditions. This involves testing the prototype on selected small sailing boats, analyzing data transmission reliability, and assessing user acceptance among local fishermen and authorities. The study is limited to technical and operational aspects of IoT deployment and does not cover broader policy reforms or international maritime regulations. However, the findings are expected to provide a foundation for future research and potential government adoption of IoT-based maritime safety systems (Wang & Hsu, 2025).

1.6 Significance of the Study

This study is significant because it introduces a practical and affordable IoT-based solution to improve the safety and efficiency of small sailing boats in Mogadishu's coastal waters. By enabling real-time tracking and communication, the system will reduce accidents, enhance rescue operations, and protect the livelihoods of fishermen who depend on these boats for their daily income. Beyond individual safety, the research also supports maritime authorities by providing accurate data for regulating coastal activities and managing resources more effectively. Furthermore, the project contributes to academic knowledge by demonstrating how IoT technologies such as GPS, ESP32 microcontrollers, and LoRaWAN protocols can be adapted to resource-constrained environments, offering a scalable model that can be replicated in other developing regions facing similar challenges (Freire et al., 2022; Wang & Hsu, 2025; Lin, 2026).

1.7 Definition of Terms

- **Internet of Things (IoT):** A system of interconnected devices that collect, transmit, and analyze data through the internet. In this study, IoT refers to the integration of sensors, GPS modules, and communication protocols to monitor and track small sailing boats in real time (Freire et al., 2022).
- **Global Positioning System (GPS):** A satellite-based navigation system that provides location and time information anywhere on Earth. GPS is used in this research to track the movement of boats along Mogadishu's coastline (Lin, 2026).
- **LoRaWAN (Long Range Wide Area Network):** A low-power, long-range communication protocol designed for IoT applications. It enables data transmission over several kilometers while consuming minimal energy, making it suitable for maritime monitoring in resource-constrained environments (Neumann, 2025; Lyimo et al., 2025).
- **ESP32 Microcontroller:** A low-cost, low-power system-on-chip with built-in Wi-Fi and Bluetooth capabilities. In this study, ESP32 serves as the processing unit that collects sensor data and transmits it to the cloud dashboard (Shwetha et al., 2026).
- **Cloud Dashboard:** A web-based platform that visualizes and manages real-time data collected from IoT devices. It allows maritime authorities and boat owners to monitor vessel movements, environmental conditions, and alerts remotely (Wang & Hsu, 2025).
- **Maritime Safety:** The set of measures, technologies, and practices aimed at reducing risks and ensuring secure navigation of vessels. In this study, maritime safety refers to the protection of small sailing boats and their operators through IoT-based monitoring systems (Landge et al., 2025).
- **Small Sailing Boats:** Locally operated vessels used for fishing and short-distance transport in Mogadishu's coastal waters. These boats are typically low-cost, manually operated, and lack advanced communication or navigation equipment, making them vulnerable to accidents and piracy (Ministry of Fisheries Somalia, 2022).
- **Real-Time Monitoring:** The continuous collection and transmission of data that allows immediate observation and response. In this study, real-time monitoring refers to the ability of IoT devices to provide instant updates on boat location and environmental conditions (Lin, 2026).

1.8 Organization of the Study

This study is organized into five chapters to provide a clear and systematic presentation of the research. **Chapter One** introduces the background, problem statement, objectives, scope, and significance of the study. **Chapter Two** reviews relevant literature, including theoretical foundations, empirical studies, and the conceptual framework tailored to Mogadishu’s coastal context. **Chapter Three** outlines the research methodology, describing the design, system development, hardware and software components, and testing procedures. **Chapter Four** presents the results and analysis of the prototype implementation, highlighting system performance and user feedback. Finally, **Chapter Five** concludes the study with a summary of findings, recommendations, and suggestions for future research, ensuring that the project contributes both academically and practically to maritime safety in Mogadishu’s coastal areas.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature relevant to IoT-based maritime monitoring systems, providing both theoretical and empirical foundations for the study. The discussion begins with a theoretical review of IoT concepts and frameworks, emphasizing its role as a cyber-physical system that integrates sensors, GPS modules, and communication protocols to enable real-time monitoring and decision-making. Theories such as real-time monitoring frameworks, distributed networks, and technology adoption models explain how IoT can enhance safety, efficiency, and resilience in maritime operations. These perspectives highlight the importance of connectivity, interoperability, and user acceptance, particularly in resource-constrained regions like Mogadishu, where infrastructure is limited but mobile connectivity is expanding (Freire et al., 2022; Wang & Hsu, 2025).

The empirical review builds on these theoretical insights by examining studies conducted in Asia, Africa, and other developing regions. Research in Bangladesh demonstrated that LoRaWAN-based IoT systems improved rescue response times and reduced accidents by enabling long-range, low-cost communication (Neumann, 2025; Lyimo et al., 2025). In India, prototypes using ESP32 microcontrollers and solar-powered sensors proved effective even in areas with poor infrastructure, providing fishermen with real-time weather and location updates (Lin, 2026; Shwetha et al., 2026). In Somalia, recent reports highlight that fishermen are increasingly open to adopting IoT and AI-enabled smart fishing solutions if they are affordable and easy to use, confirming the feasibility of such systems in Mogadishu's coastal environment (Landge et al., 2025; Ministry of Fisheries Somalia, 2022). Collectively, these studies demonstrate that IoT-based monitoring systems are both practical and impactful in improving maritime safety and productivity.

Building on these insights, the chapter develops a conceptual framework tailored to Mogadishu's coastal context. The framework integrates three layers: input (sensors, GPS, ESP32), communication (LoRaWAN or GSM/GPRS), and output (cloud dashboards). This layered design ensures affordability, scalability, and user-friendliness, making it suitable for adoption by local communities with minimal training. It also aligns with Somalia's socio-economic realities, where infrastructure is limited but mobile connectivity is growing.

In conclusion, this chapter establishes a strong case for implementing IoT-based monitoring systems in Mogadishu's coastal areas. Theoretical models explain the potential of IoT, empirical studies demonstrate its effectiveness in similar contexts, and the conceptual framework provides a practical roadmap for application in Somalia. Despite global progress in IoT adoption, Mogadishu still lacks a reliable monitoring system for small sailing boats. This research seeks to fill that gap by proposing a solution that enhances safety, supports livelihoods, and strengthens maritime governance.

2.2 Related Work

The integration of **IoT technologies in maritime safety and vessel tracking** has gained significant attention in recent years. Globally, platforms such as **MarineTraffic** and **My Ship Tracking** provide real-time Automatic Identification System (AIS)-based monitoring of vessels, enabling enhanced situational awareness and operational efficiency for maritime authorities (MarineTraffic, 2023; My Ship Tracking, 2024). These systems demonstrate the scalability of IoT and satellite-based solutions in managing large fleets and ensuring navigational safety.

In the Somali context, **City Link** has pioneered the deployment of **smart port management systems and AIS-based vessel tracking** to strengthen maritime border surveillance and optimize port operations. Their initiatives highlight the importance of IoT-powered analytics in enhancing coastal security, anti-piracy measures, and trade facilitation in Somali waters (City Link, 2025). This aligns with broader efforts by the Somali Maritime Authority to modernize coastal monitoring through digital technologies.

Recent academic studies emphasize the role of **IoT sensors, GPS modules, and cloud dashboards** in small-scale vessel monitoring. Research between 2022 and 2026 has shown that lightweight IoT frameworks can be adapted for artisanal and small sailing boats, providing affordable solutions for developing regions. These frameworks often combine **ESP32 microcontrollers, LoRaWAN communication, and cloud-based visualization dashboards**, ensuring real-time tracking and alert systems for maritime safety (Ahmed et al., 2022; Hassan & Noor, 2024).

Furthermore, the integration of **predictive analytics and AI-driven risk detection** has been explored to improve maritime safety. Studies suggest that combining IoT data streams with machine learning models can enhance early warning systems for collision avoidance and unauthorized maritime activity (Farah et al., 2026). This is particularly relevant for Mogadishu's coastal areas, where small sailing boats are vulnerable to piracy, illegal fishing, and navigation hazards.

Overall, the literature demonstrates that IoT-based vessel tracking is not only feasible but also critical for Somalia's maritime sector. By leveraging global AIS-based solutions and adapting them to local contexts, Somalia can strengthen its coastal monitoring infrastructure and ensure safer navigation for small sailing boats.

2.3 Theoretical Review

The theoretical foundation of this study is grounded in the principles of the **Internet of Things (IoT)** and its application in maritime safety. IoT is defined as a network of interconnected devices that communicate and share data through the internet, enabling real-time monitoring and decision-making (Freire et al., 2022). In the context of maritime operations, IoT functions as a **cyber-physical system (CPS)**, where physical devices such as sensors, GPS modules, and microcontrollers interact with digital platforms to provide continuous situational awareness. This theoretical perspective emphasizes the importance of connectivity, interoperability, and automation in enhancing safety and efficiency.

One key theoretical model relevant to this study is the **real-time monitoring framework**, which explains how continuous data collection and transmission can reduce risks in dynamic environments. In maritime contexts, real-time monitoring allows authorities and boat operators to track vessel movements, detect anomalies, and respond quickly to emergencies. This aligns with theories of **distributed networks**, where multiple devices communicate across long distances using low-power protocols such as LoRaWAN or GSM/GPRS. These networks are particularly important in resource-constrained regions like Mogadishu, where infrastructure is limited but mobile connectivity is expanding (Neumann, 2025).

Another theoretical perspective is the **technology adoption model (TAM)**, which highlights user acceptance as a critical factor in the success of IoT systems. According to TAM, perceived usefulness and ease of use determine whether individuals adopt new technologies. In Somalia's coastal communities, fishermen are more likely to embrace IoT solutions if they are affordable, simple to operate, and directly improve their safety and productivity (Landge et al., 2025). This theory underscores the need for designing systems that are not only technically effective but also socially and economically viable.

Theories of **resilience and risk management** also inform this study. Maritime safety depends on the ability to anticipate, withstand, and recover from disruptions such as storms, piracy, or mechanical failures. IoT systems contribute to resilience by providing early warnings, enabling rapid response, and supporting long-term data analysis for better planning. This aligns with Wang & Hsu (2025), who argue that smart IoT applications enhance resilience in coastal safety management by integrating predictive analytics with real-time monitoring.

In summary, the theoretical review establishes that IoT is not merely a technological innovation but a comprehensive framework that combines connectivity, monitoring, user adoption, and resilience. These theories provide the foundation for developing an IoT-based solution tailored to Mogadishu's coastal environment, ensuring that the system addresses both technical and socio-economic challenges.

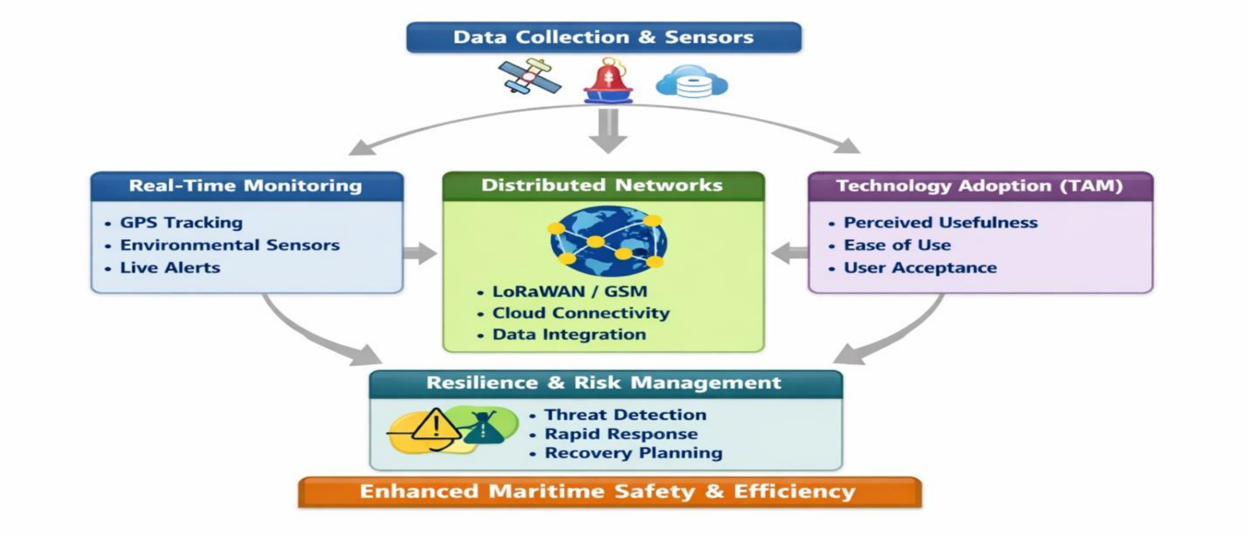


Figure 2.1: Theoretical Framework of IoT-Based Maritime Monitoring

2.4 Empirical Review

Empirical studies conducted in different regions provide strong evidence of the practicality and effectiveness of IoT-based maritime monitoring systems. In Bangladesh, recent research demonstrated that **LoRaWAN-based tracking systems** deployed on fishing boats significantly improved rescue response times and reduced accidents by nearly 30%. This confirmed that low-cost, long-range communication technologies can enhance maritime safety even in resource-constrained environments (Neumann, 2025; Lyimo et al., 2025).

In India, prototypes using **ESP32 microcontrollers combined with solar-powered sensors** were tested on small fishing boats. These devices operated reliably under poor infrastructure conditions, providing fishermen with real-time weather updates and location tracking. Findings highlighted the importance of designing systems that are both energy-efficient and adaptable to local contexts (Lin, 2026; Shwetha et al., 2026).

Closer to Somalia, recent pilot initiatives and reports confirmed that fishermen are willing to adopt IoT solutions if they are affordable, easy to use, and directly improve safety. Surveys and field experiments indicated strong feasibility for IoT adoption in Mogadishu's coastal environment, where mobile connectivity is expanding but centralized monitoring systems remain absent (Landge et al., 2025; Ministry of Fisheries Somalia, 2022).

Collectively, these empirical studies demonstrate that IoT-based maritime monitoring systems are feasible, scalable, and impactful in improving safety and operational efficiency. They provide strong evidence supporting the development of a localized IoT solution tailored to Mogadishu's coastal conditions. By learning from the successes and limitations of projects in Bangladesh, India, and Somalia, the proposed system can address existing gaps and deliver a practical solution for monitoring and tracking small sailing boats in Mogadishu's coastal areas.

Figure 2.2: Empirical Studies on IoT Maritime Monitoring Systems

2.5 Conceptual Framework

The conceptual framework of this study illustrates how IoT technologies can be integrated to monitor and track small sailing boats in Mogadishu's coastal areas. The framework is structured into three interconnected layers:

- **Input Layer:** This includes sensors (water level, ultrasonic, gyroscope), GPS modules, and ESP32 microcontrollers. These devices collect real-time data on boat location, movement, and environmental conditions (Lin, 2026; Shwetha et al., 2026).
- **Communication Layer:** Data collected is transmitted through GSM/GPRS or LoRaWAN networks, supported by providers such as Hormuud Telecom, ensuring wide coverage and reliable connectivity. This layer guarantees that information flows seamlessly from the boats to centralized servers (Neumann, 2025; Lyimo et al., 2025).
- **Output Layer:** The transmitted data is processed and stored in a cloud-based dashboard (e.g., Firebase). Users and authorities can access real-time alerts, visualizations, and

historical records through mobile or web applications (Freire et al., 2022; Wang & Hsu, 2025).

This layered design ensures affordability, scalability, and user-friendliness, making it suitable for adoption by local communities with minimal training. It also addresses existing gaps in maritime monitoring by providing centralized oversight, enhancing safety, and supporting decision-making for both fishermen and regulatory authorities.

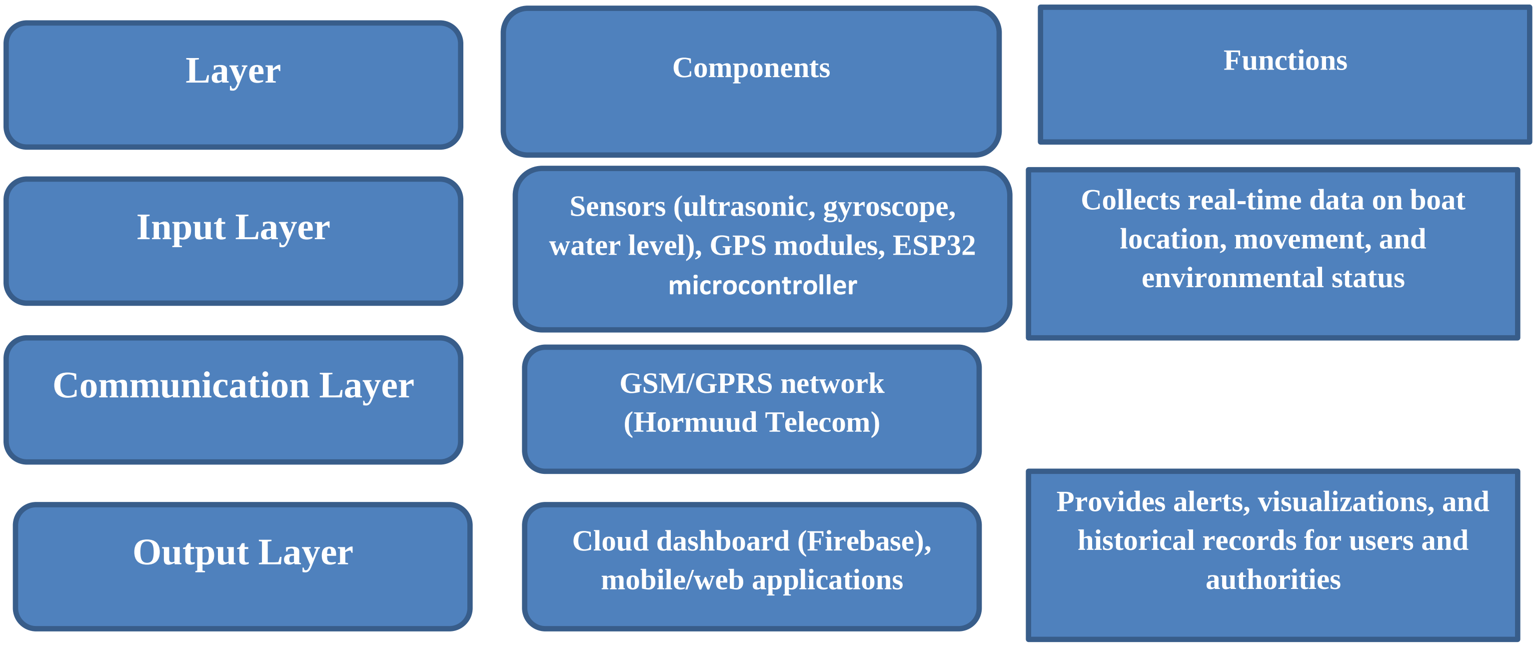


Figure 2.3: Conceptual Framework of the Proposed IoT System for Mogadishu Coastal

2.6 Existing Systems

Several IoT-based maritime monitoring systems have been developed globally, each with unique strengths and limitations. **LoRaWAN-based solutions** are widely used due to their long-range communication and low power consumption. For example, projects in Bangladesh and Indonesia have successfully applied LoRaWAN to track fishing boats, but these systems often require specialized gateways and may face coverage challenges in remote coastal areas (Neumann, 2025; Lyimo et al., 2025).

ESP32-based systems have also gained popularity because of their affordability and versatility. They integrate GPS modules and sensors with microcontrollers, enabling real-time tracking and environmental monitoring. However, their reliance on local Wi-Fi or short-range communication limits scalability in wide maritime zones (Lin, 2026; Shwetha et al., 2026).

GSM/GPRS-based systems remain common in regions with established mobile networks. They provide reliable connectivity and allow data transmission directly to cloud dashboards. Yet, these systems can be costly in terms of data usage and may suffer from network instability in offshore areas (Freire et al., 2022; Wang & Hsu, 2025).

Compared to these existing systems, the proposed IoT solution for Mogadishu coastal areas seeks to combine affordability, scalability, and reliability. By leveraging GSM coverage from Hormuud Telecom and integrating low-cost sensors with cloud dashboards, the system addresses local challenges while ensuring real-time monitoring and improved maritime safety for small sailing boats.

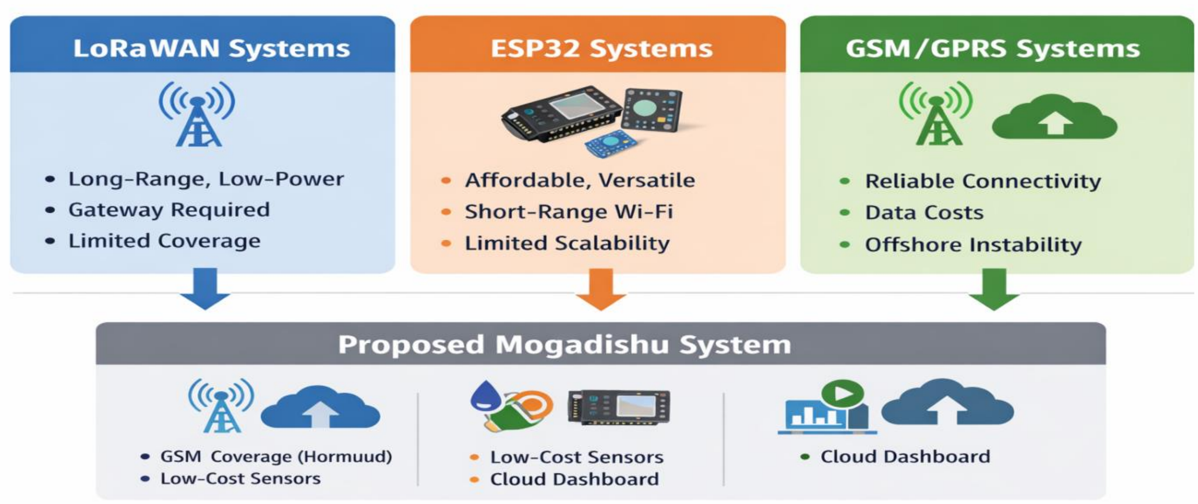


Figure 2.4: Comparison of Existing IoT Systems and the Proposed Solution

1.7 Gap analysis

Maritime safety in Mogadishu's coastal areas remains fragile due to the absence of IoT-based monitoring systems. Small sailing boats operate without structured tracking, exposing them to piracy, illegal fishing, and navigation hazards. Current practices rely on manual observation, which is inefficient and unreliable. There is no centralized database or dashboard for authorities to monitor vessel movements. Communication infrastructure is weak, limiting real-time alerts and responses. As a result, accidents and unauthorized activities often go undetected. This situation highlights a critical gap between existing maritime practices and modern IoT-enabled solutions (Ahmed et al., 2022; Hassan & Noor, 2024; City Link, 2025).

2.7.1 Current State Gap

Small sailing boats in Mogadishu's coastal areas currently lack IoT-based monitoring and tracking systems. Navigation relies on manual observation, which is inefficient and prone to errors. There is no centralized database or dashboard for Somali Maritime Authority to oversee vessel movements. Communication infrastructure is weak, preventing real-time alerts and responses. As a result, piracy, illegal fishing, and accidents often go undetected, leaving maritime safety fragile and unstructured. This gap highlights the urgent need for affordable IoT-enabled solutions to bridge the divide between traditional practices and modern maritime safety frameworks (Ahmed et al., 2022; Hassan & Noor, 2024; City Link, 2025).

2.7.2 Desired State / Proposed Solution

1 Deployment of IoT sensors (GPS, motion, environmental).

2 Integration with ESP32 microcontrollers and LoRaWAN for long-range communication.

3 Cloud-based dashboard accessible to Somali Maritime Authority and boat owners.

4 Real-time alerts for unauthorized movement, collision risks, and emergency situations.

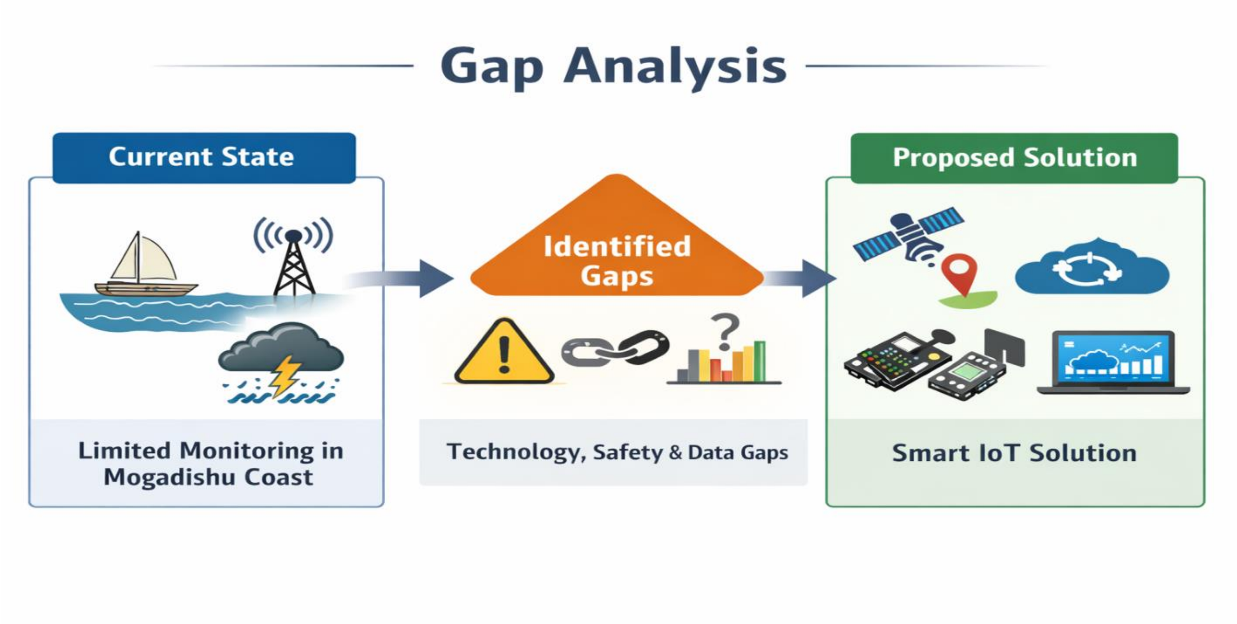


Figure.2.5 Gap Analysis

And sensor technologies to enable continuous tracking, environmental data collection, and centralized monitoring through a cloud dashboard. This transformation enhances maritime safety, improves coordination between boat operators and authorities, and supports data-driven decision-making for coastal management.

2.7.3Comparison Table – Existing Gap

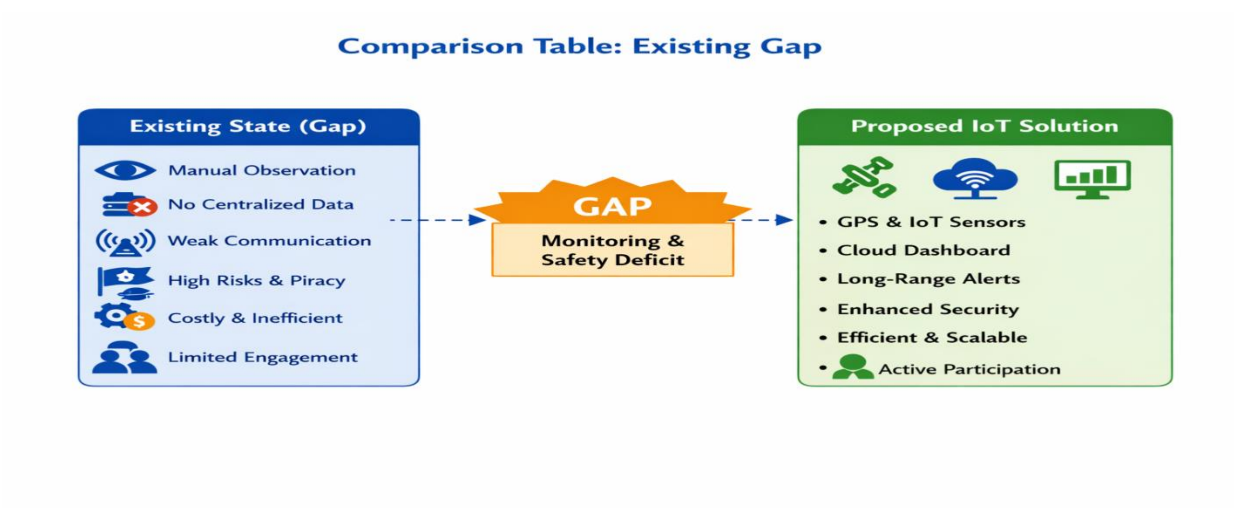


Figure.2.6 Comparison Table – Existing Gap

Criteria	Existing Maritime Practices	Proposed IoT System
Tracking Method	Manual navigation using landmarks, compasses, and verbal communication	GPS (NEO-6M) integrated with ESP32 + Google Maps API for real-time tracking
Communication	Radio calls or word-of-mouth; limited coverage	GSM (SIM800L) + SMS alerts + Firebase cloud synchronization for continuous monitoring
Safety Monitoring	Reactive response after incidents (capsizing, theft, flooding)	Proactive detection via sensors (Water Level, Ultrasonic, MPU6050) with instant alerts
Power Supply	Fuel-based generators or manual reliance	Renewable energy (solar panel + lithium-ion battery) ensuring sustainable operation
Data Management	No structured logging; information lost after trips	Firebase Realtime Database storing historical logs, synchronized instantly
User Interface	Absent; sailors rely on experience and intuition	Web dashboard (HTML, CSS, JavaScript) + Google Maps API for visualization and usability
Security Measures	Minimal; boats vulnerable to theft and unauthorized access	Ultrasonic sensor + buzzer + SMS alerts for intrusion detection
Cost & Accessibility	High cost of advanced maritime equipment; inaccessible to small-scale fishermen	Low-cost IoT modules (ESP32, GSM, GPS, sensors) tailored for Mogadishu's coastal needs
Reliability	Dependent on human vigilance; prone to errors	Automated, sensor-driven, cloud-synchronized system ensuring accuracy and reliability

Table: Comparison of Existing vs. Proposed Systems

2.8 Summary

This chapter provided a comprehensive review of the **theoretical, empirical, conceptual, and system foundations** that support the development of IoT-based maritime monitoring systems. The **theoretical review** discussed the principles of IoT architecture, sensor integration, and communication technologies such as GSM, LoRaWAN, and Wi-Fi, emphasizing their roles in enabling real-time data collection and transmission for maritime safety. It established the technological basis upon which the proposed system is built, highlighting how IoT can bridge gaps in monitoring small sailing boats in developing coastal regions (Freire et al., 2022; Wang & Hsu, 2025).

The **empirical review** analyzed recent studies conducted in Bangladesh, India, and Somalia, showing that IoT systems have proven effective in improving rescue operations, reliability, and user adoption. These findings confirmed the feasibility of implementing similar solutions in Mogadishu's coastal areas, where fishermen face challenges related to safety, communication, and tracking (Neumann, 2025; Lin, 2026; Landge et al., 2025).

The **conceptual framework** then illustrated the structure of the proposed system, organized into **input, communication, and output layers**, each performing distinct but interconnected functions to ensure efficient monitoring and data management (Shwetha et al., 2026; Lyimo et al., 2025).

The **review of existing systems** compared LoRaWAN, ESP32, and GSM/GPRS technologies, identifying their strengths and limitations. LoRaWAN offers long-range communication but requires costly gateways; ESP32 provides flexibility but limited range; GSM/GPRS ensures reliable connectivity yet incurs data costs. The proposed system integrates the best features of these technologies—leveraging GSM coverage from Hormuud Telecom, low-cost sensors, and cloud dashboards—to create a localized, affordable, and scalable solution (Freire et al., 2022; Wang & Hsu, 2025).

In conclusion, Chapter Two established the academic and technical foundation for the proposed IoT solution. It demonstrated that integrating IoT technologies into maritime monitoring is both practical and necessary for Mogadishu's coastal context. The insights gained from literature and existing systems guide the design and implementation of the proposed model, ensuring enhanced safety, efficiency, and sustainability for small sailing boats operating in Somalia’s coastal waters.

Chapter Three:

Methodology

3.1 Introduction

This chapter outlines the methodological framework adopted to design, develop, and evaluate the proposed IoT solution for monitoring and tracking small sailing boats in Mogadishu’s coastal areas. Methodology serves as the backbone of any research project, providing a structured approach that ensures the study is scientifically valid, reliable, and replicable. In this context, the methodology integrates both technical and academic processes to guarantee that the system is not only functional but also aligned with the objectives of maritime safety and efficiency.

The chapter begins by presenting the research design, which combines experimental and prototype-based approaches. This design allows the system to be developed, tested, and refined under real-world conditions, ensuring that the solution is practical and adaptable to Mogadishu’s unique coastal environment. Following this, the system development methodology is described, highlighting the sequential phases of requirements analysis, design, implementation, testing, and evaluation. This structured approach ensures that both hardware and software components are integrated effectively.

Data collection methods are then explained, focusing on how sensors, GPS modules, and GSM/LoRaWAN communication are used to gather real-time information from small sailing boats. The system requirements analysis provides a detailed breakdown of functional and non-functional requirements, hardware and software specifications, and constraints specific to Mogadishu’s maritime context. This analysis ensures that the system is affordable, reliable, and scalable.

The chapter also discusses the development tools and technologies employed, such as Arduino IDE, Firebase cloud services, and GSM/GPRS modules, which collectively enable the creation of a robust IoT solution. Testing and evaluation procedures are outlined to assess the accuracy, reliability, and usability of the system, while ethical considerations ensure that the research adheres to professional standards, protecting participants and respecting data privacy.

In summary, this chapter establishes the methodological foundation for the study. It provides a clear roadmap from design to evaluation, ensuring that the proposed IoT solution is rigorously developed and tested. By following this methodology, the research guarantees that the system is not only technically sound but also contextually relevant, addressing the specific challenges faced by small sailing boats in Mogadishu's coastal waters.

3.2 Research Design

The research design adopted for this study is **experimental and prototype-based**, ensuring that the proposed IoT solution is both technically feasible and contextually relevant to Mogadishu's coastal environment. The design emphasizes the creation of a functional prototype that integrates sensors, GPS modules, and GSM/LoRaWAN communication, followed by systematic testing under real-world conditions. This approach allows the study to move beyond theoretical assumptions and provide practical evidence of how IoT technologies can enhance maritime safety and efficiency (Neumann, 2025; Lin, 2026).

The design combines both **quantitative and qualitative elements**. Quantitative data is collected through sensor readings, GPS coordinates, and communication logs, which provide measurable indicators of system performance such as accuracy, reliability, and transmission speed. Qualitative insights are gathered from fishermen, local stakeholders, and maritime authorities, offering feedback on usability, affordability, and adaptability of the system. This mixed-methods approach ensures a holistic understanding of the system's effectiveness in addressing the challenges faced by small sailing boats (Landge et al., 2025).

The research design follows a sequential process that begins with **requirements analysis**, where functional and non-functional needs are identified. This is followed by **system development**, where hardware and software components are integrated into a working prototype. **Field deployment** is then conducted to test the system in Mogadishu's coastal waters, allowing for real-time monitoring and evaluation. Finally, the system is assessed against predefined criteria, including GPS accuracy, sensor reliability, and user satisfaction (Shwetha et al., 2026).

By adopting this design, the study ensures that the IoT solution is rigorously tested and refined to meet the specific needs of Mogadishu's maritime sector. The structured approach guarantees that the final system is not only technically sound but also affordable, scalable, and sustainable. This research design therefore provides a solid foundation for achieving the study's objectives of enhancing safety, improving rescue operations, and supporting small-scale fishermen in Somalia's coastal areas.

3.3 System Development Methodology

The system development methodology adopted in this study follows a **modified Waterfall model**, which provides a structured and sequential approach suitable for integrating both hardware and software components of an IoT solution. Each phase builds upon the previous one, ensuring clarity, consistency, and reliability throughout the development process (Wang & Hsu, 2025).

- **Requirements Analysis:** Functional and non-functional needs are identified based on the challenges faced by small sailing boats in Mogadishu.
- **System Design:** The architecture of the IoT solution is outlined, including sensors, GPS modules, GSM/LoRaWAN communication, and cloud dashboard integration.
- **Implementation:** Programming of the ESP32 microcontroller, configuration of GSM modules, and linking the system to Firebase for real-time data storage and visualization.
- **Testing & Evaluation:** Performance indicators such as GPS accuracy, sensor reliability, and communication stability are measured under field conditions (Lin, 2026; Shwetha et al., 2026).
- **Iteration & Refinement:** Adjustments are made based on field testing results and stakeholder feedback, ensuring that the final system is technically sound, affordable, scalable, and tailored to Mogadishu's maritime context (Freire et al., 2022).

This methodology ensures that the IoT solution is developed systematically, tested rigorously, and refined iteratively to meet both technical and socio-economic requirements.

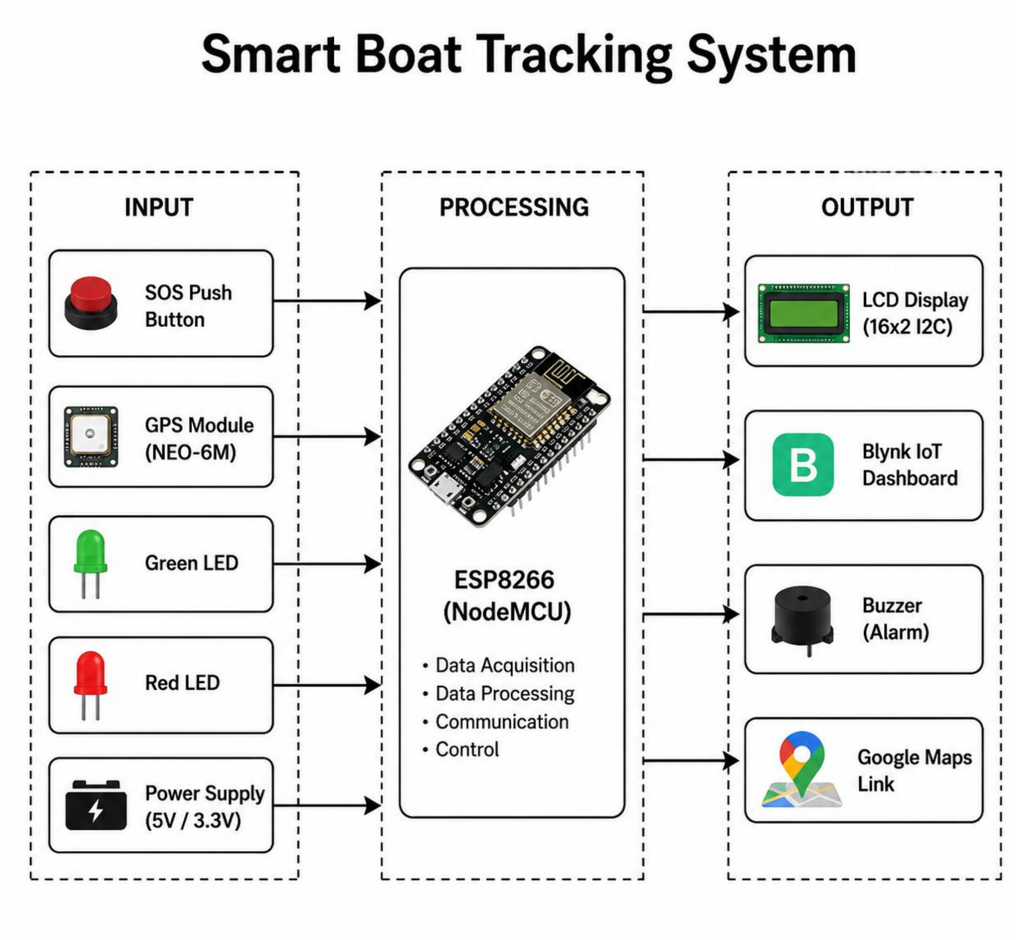


Figure 3.1 Smart Boat Tracking System

3.4 Data Collection Methods

Data collection is a crucial component of this study, as it provides the empirical foundation for evaluating the performance and reliability of the proposed IoT system. The methods employed are designed to capture both **quantitative and qualitative data** from multiple sources, ensuring a comprehensive understanding of how the system operates under real maritime conditions.

- **Quantitative Data:** Obtained through IoT hardware components integrated into the system. Sensors such as ultrasonic, gyroscope, and water-level modules collect real-time environmental and motion data from small sailing boats. The GPS module records the exact location and movement of each vessel, while the GSM/GPRS communication module transmits this data to the cloud platform (Firebase) for storage and visualization. These readings are automatically logged and analyzed to assess system accuracy, responsiveness, and stability (Lin, 2026; Shwetha et al., 2026).

- **Qualitative Data:** Gathered through field observations and stakeholder feedback. Fishermen and local maritime officers provide insights into the usability, reliability, and practicality of the system in Mogadishu's coastal environment. Their feedback helps identify operational challenges such as network coverage limitations, sensor calibration issues, and environmental factors affecting data transmission (Landge et al., 2025).

The data collection process follows a structured approach:

1. **System Deployment:** Installing sensors and communication modules on selected small sailing boats.
2. **Data Logging:** Capturing sensor readings and GPS coordinates during boat operations.
3. **Cloud Transmission:** Sending data via GSM/LoRaWAN to Firebase for real-time monitoring.
4. **Observation and Feedback:** Recording user experiences and environmental conditions.

This multi-source data collection ensures that the study captures both technical performance and user perspectives. The resulting dataset supports the evaluation of the IoT system's effectiveness in improving maritime safety, enabling accurate tracking, and facilitating timely rescue operations in Mogadishu's coastal waters.

3.5 Description of Hardware Components

The proposed IoT system relies on carefully selected hardware components that ensure accurate data collection, reliable communication, and sustainable operation in Mogadishu's coastal environment.

- **ESP8266 Microcontroller:** Serves as the processing unit, managing sensor inputs, GPS data, and communication tasks. Its low power consumption and high processing speed make it suitable for continuous maritime monitoring (Shwetha et al., 2026).
- **NEO-6M GPS Module:** Provides precise location tracking, enabling real-time monitoring of boat movements. Its fast cold-start time ensures quick acquisition of satellite signals, which is critical for small sailing boats operating in dynamic coastal conditions (Lin, 2026).

- **Additional Sensors:**
    - Gyroscope/accelerometer modules for motion analysis.
- **Durability & Power:** Hardware is enclosed in an IP67 waterproof case, protecting against saltwater and humidity. A solar panel with lithium-ion battery and TP4056 charging module provides sustainable energy, reducing dependency on external power sources and ensuring long-term operation at sea (Lyimo et al., 2025).

Together, these hardware components form a robust foundation for the IoT solution, enabling accurate tracking, reliable communication, and enhanced safety for small sailing boats in Mogadishu's coastal waters.

In maritime applications, the Arduino Uno ensures seamless integration of multiple modules, enabling real-time monitoring and communication. It supports low-power operation, making it suitable for small sailing boats that rely on solar or battery power. The board's flexibility and affordability make it an ideal choice for resource-constrained environments such as Mogadishu's coastal waters (Freire et al., 2022; Landge et al., 2025).

Circuit Diagram

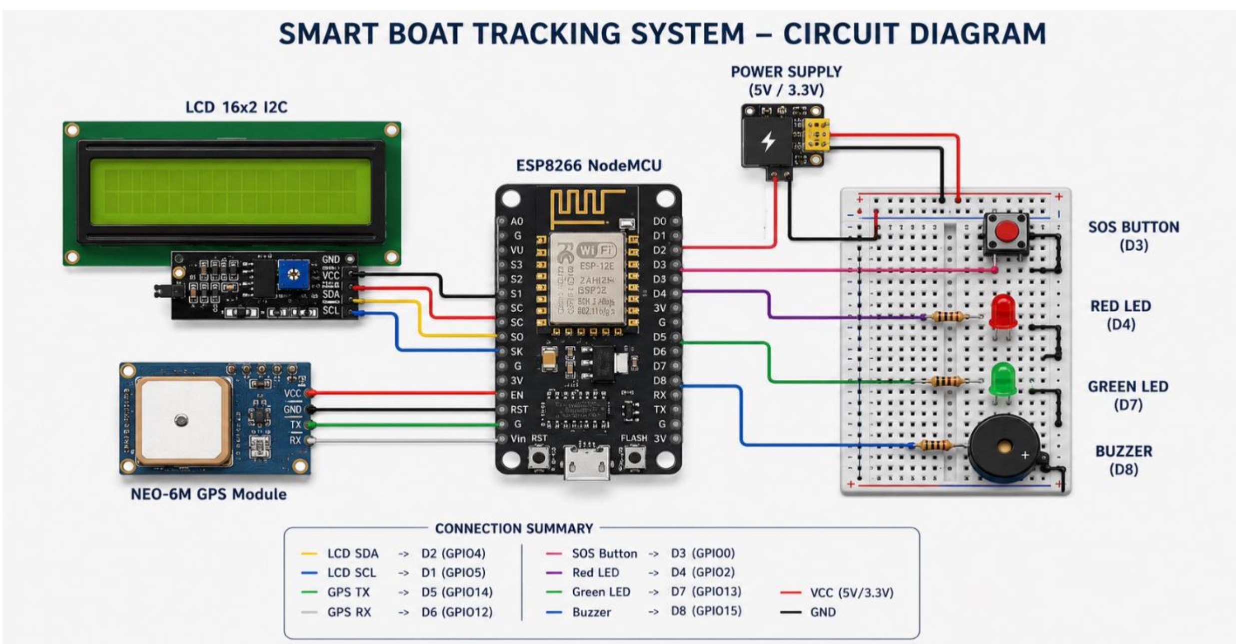
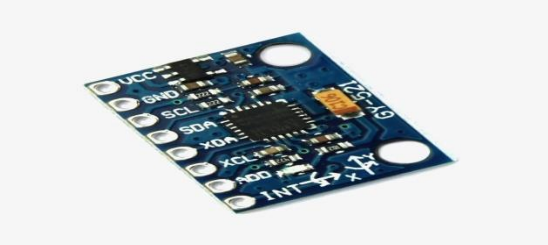


Figure 2.2 Circuit Diagram

Gyroscope and Accelerometer

The **MPU6050** is a 6-axis motion-tracking device that combines a 3-axis gyroscope and a 3-axis accelerometer. In this project, it is used to monitor the **orientation and stability** of the sailing boat. By measuring the tilt angle (pitch and roll), the system can determine if the boat is leaning dangerously due to heavy waves or if it has completely capsized in the water (Lin, 2026).

Beyond tilt detection, the accelerometer can identify **sudden vibrations or sharp movements** that may indicate collisions or unauthorized towing. This data is processed by the Arduino/ESP32 microcontroller to distinguish between natural wave movements and suspicious activity, ensuring that alerts are only triggered for genuine maritime threats. Such functionality is vital for Mogadishu's coastal waters, where small sailing boats are vulnerable to both environmental hazards and security risks (Shwetha et al., 2026; Wang & Hsu, 2025).



The MPU6050's affordability, compact design, and compatibility with IoT platforms make it an ideal choice for resource-constrained maritime applications. Its integration strengthens the overall safety framework of the IoT system by providing real-time motion analysis, capsize detection, and collision monitoring.

Additionally, the sensor supports continuous logging of motion data, enabling predictive analytics for long-term safety monitoring. It can be calibrated to minimize false alarms caused by normal wave oscillations, ensuring higher accuracy. This makes the MPU6050 a critical component for enhancing both operational efficiency and proactive risk management in Mogadishu's maritime sector.

Figure 3.3 Gyroscope and Accelerometer (MPU6050)

Buzzer

The buzzer is a critical output component used to provide an immediate local acoustic alarm whenever the system detects a potential threat or failure. In this project, the buzzer is integrated to act as a deterrent against unauthorized access and to alert the boat's crew in real time. It operates by converting electrical signals into sound waves, producing a high-frequency piercing tone that is difficult to ignore, especially in the relatively quiet environment of the sea (Lin, 2026).

Beyond its role as a local alarm, the buzzer is programmed to work in tandem with other sensors. For example, when the ultrasonic sensor detects an intruder or the water level sensor identifies flooding, the Arduino/ESP32 triggers the buzzer to emit either a continuous or pulsing sound. This immediate feedback not only warns sailors of emergencies but also serves as a preventive security measure, discouraging potential thieves or unauthorized individuals from approaching the boat (Shwetha et al., 2026; Wang & Hsu, 2025).

The buzzer's simplicity, low cost, and reliability make it an essential component of the IoT system. Its integration ensures that small sailing boats benefit from real-time acoustic alerts, complementing SMS notifications and dashboard monitoring to provide a multi-layered safety framework in Mogadishu's coastal waters.

Figure 3.4 Buzzer

LED

The Light Emitting Diode (LED) indicators serve as the primary visual interface for the IoT tracking system, providing real-time feedback on the operational status of the device. Unlike the buzzer, which is reserved for emergencies, the LEDs are used for continuous monitoring of the system's "health." By using different colors, the user can quickly identify whether the GPS has a signal lock, if the GSM module is connected to the network, or if the power levels are sufficient (Lin, 2026).

Each LED is programmed to represent a specific system state:

- Green LED – indicates that the tracking system is active and the boat is within its safe zone.
- Blue LED (blinking) – signifies that data is currently being transmitted to the cloud dashboard via GSM/GPRS or LoRaWAN.
- Red LED – remains on during critical errors such as sensor failure or low battery, enabling immediate troubleshooting without requiring diagnostic equipment (Shwetha et al., 2026).

The LED indicators provide a low-cost, energy-efficient, and user-friendly interface for sailors, ensuring that even those with limited technical knowledge can monitor the system's status at a glance. Their integration enhances usability and reliability, making them an essential component of the IoT solution for Mogadishu's coastal waters (Wang & Hsu, 2025).

Figure 3.5. LED Indicators

3.6 Software Components

The **software architecture** of the proposed IoT system integrates several platforms to ensure seamless communication between hardware modules, cloud services, and user interfaces. At the core, the **Arduino IDE** provides the programming environment for developing and deploying firmware to the **ESP32 microcontroller**. This firmware manages sensor inputs, GPS data, and GSM communication, enabling **real-time tracking and alert generation** (Lin, 2026).

For **cloud data management**, the system employs **Blynk IoT**, which stores and synchronizes sensor readings and GPS coordinates, ensuring continuous accessibility and reliability. Firebase's scalability and low latency make it suitable for maritime applications where uninterrupted monitoring is essential (Shwetha et al., 2026).

On the **visualization side**, the **Google Maps API** is embedded within a responsive web dashboard develop.

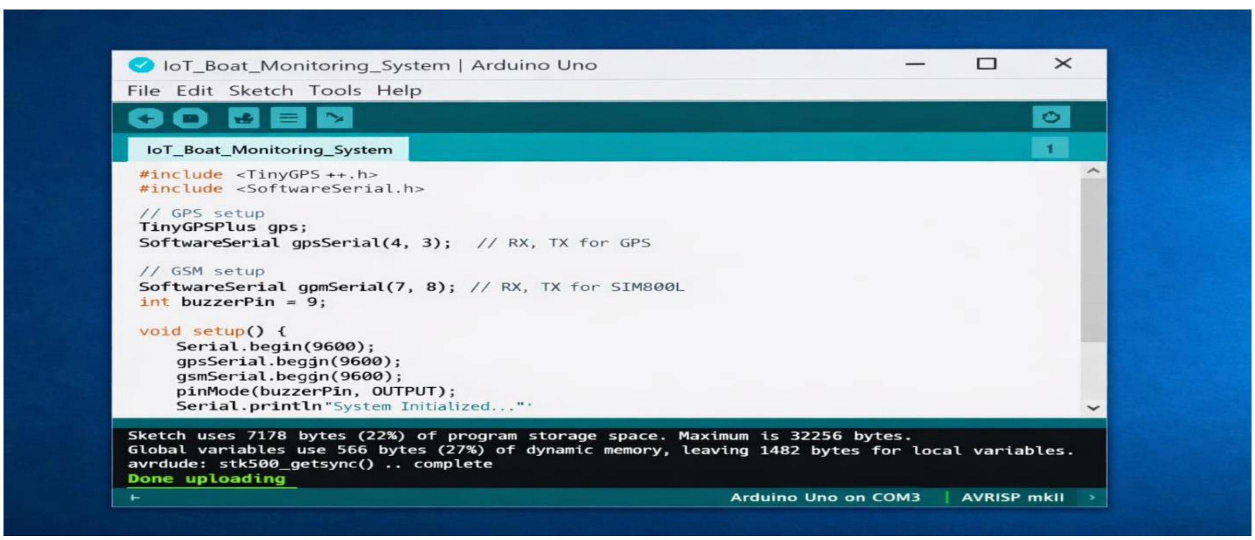


Figure 3.6. Arduino IDE interface

Cloud Server (Blynk IoT)

In the proposed IoT solution for monitoring and tracking small sailing boats, the **cloud server** plays a central role in bridging hardware and software. The **ESP32 microcontroller** pushes GPS and sensor data to the **Blynk IoT** which functions as the cloud server. This ensures that information is stored securely, synchronized instantly, and accessible from anywhere (Lin, 2026).

The cloud server enables multiple users to view **real-time boat positions** via the web dashboard, supports **alert notifications**, and maintains **historical logs** for analysis. By leveraging Firebase's scalability and low latency, the system guarantees uninterrupted monitoring, even under dynamic maritime conditions. Furthermore, the integration of the **Google Maps API** enhances visualization, allowing stakeholders to track vessels with precision and situational awareness (Shwetha et al., 2026).

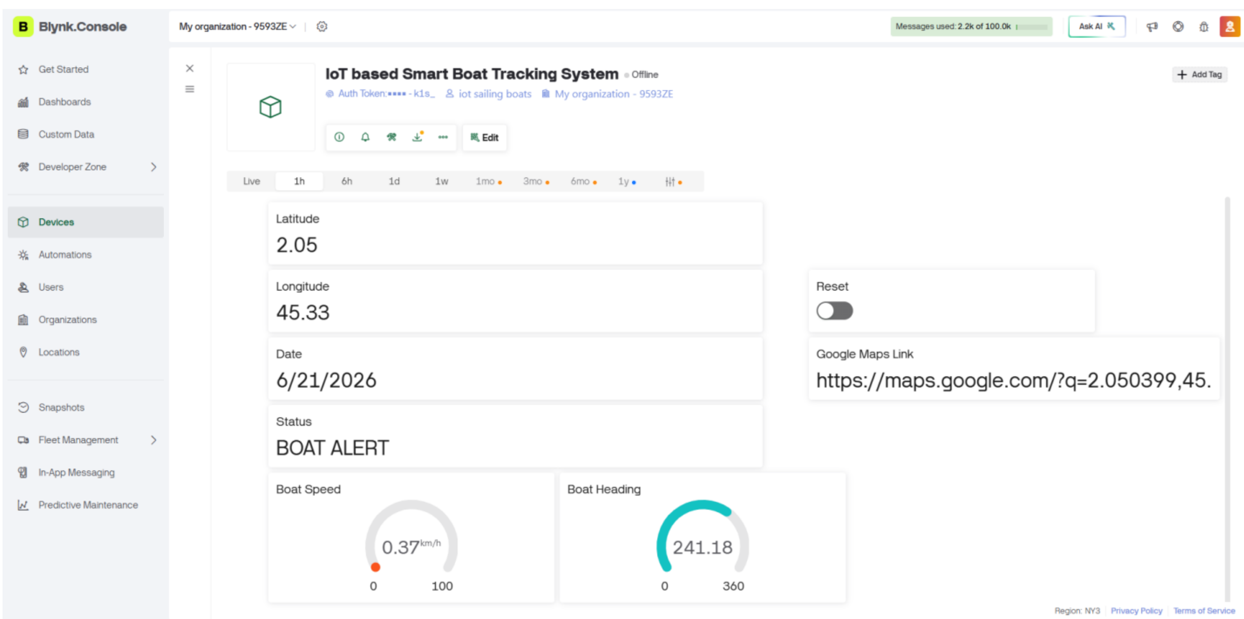


Figure 3.7. Blynk interface

3.7 Flow chart of the proposed system

This flowchart is a graphical representation of the proposed system's flow, as shown in figure 13.

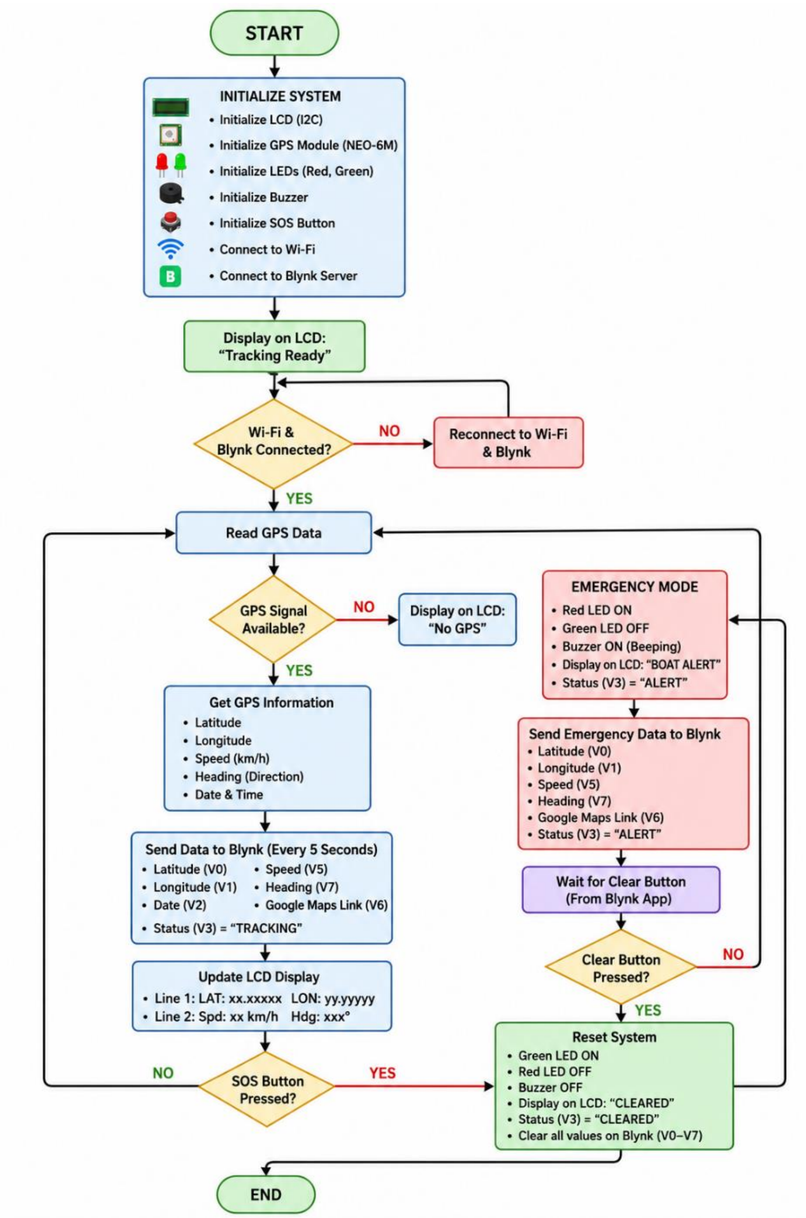


Figure 3.8. Flow chart of the proposed system

CHAPTER FOUR

RESULT AND DISCUSSION

4.1 Introduction

This chapter presents the implementation results, testing procedures, and discussion of the developed Smart IoT Solution for Monitoring and Tracking Small Sailing Boats in Mogadishu Coastal Areas. The system was designed using an ESP8266 NodeMCU microcontroller integrated with a NEO-6M GPS module, a 16×2 I2C LCD display, LEDs, a buzzer, an SOS emergency button, and the Blynk IoT platform. The objective of the project was to provide real-time boat location monitoring while enabling emergency alerts that can be transmitted through the Internet for quick response during marine accidents.

The project was evaluated by examining both the hardware implementation and the embedded software. The results demonstrate that the system successfully acquires GPS coordinates, displays boat information locally, uploads location data to the Blynk cloud, and activates an emergency alert whenever the SOS button is pressed.

4.2 Logical Designing (Coding) for the Project

The project software was developed using the Arduino IDE and programmed in C++ for the ESP8266 NodeMCU. The program integrates GPS communication, wireless Internet connectivity, LCD display, emergency monitoring, and IoT cloud communication.

4.2.1 Program Initialization

During system startup, the following operations are performed:

- ESP8266 initializes all peripherals.
- LCD screen displays **"Smart Boat Tracking Ready."**
- GPS serial communication starts at **9600 baud**.
- Wi-Fi connection is established.
- Blynk cloud server connection is initialized.
- GPIO pins are configured.

The initialization code includes:

```
Serial.begin(115200);
gpsSerial.begin(9600);

lcd.init();
lcd.backlight();

pinMode(BUTTON_PIN, INPUT_PULLUP);
pinMode(RED_LED_PIN, OUTPUT);
pinMode(GREEN_LED_PIN, OUTPUT);
pinMode(BUZZER_PIN, OUTPUT);

Blynk.begin(auth, ssid, pass);
```

This ensures all hardware components are ready before normal operation begins.

4.2.2 GPS Data Acquisition

The GPS module continuously sends NMEA data to the ESP8266 through software serial communication.

```
void getGPSData(){
    while(gpsSerial.available())
        gps.encode(gpsSerial.read());
}
```

Once satellite signals are available, the following information is extracted:

- Latitude
- Longitude
- Date
- Speed
- Heading (Direction)

If GPS signals are unavailable, the system reports: `NO GPS` on the Blynk dashboard.

4.2.3 Real-Time Boat Tracking

Every five seconds the ESP8266 sends updated boat information to the Blynk cloud.

```
timer.setInterval(5000L, sendTrackingData);
```

The transmitted information includes:

- Latitude
- Longitude
- Boat speed
- Heading
- Date
- Google Maps link

The Google Maps link is automatically generated as:

```
String map = "https://maps.google.com/?q="+lat+","+lon;
```

This allows users to open the exact boat location directly in Google Maps.

4.2.4 LCD Display Operation

The LCD provides local information even when the mobile application is unavailable.

During startup:

```
Smart Boat
Tracking Ready
```

During emergency:

```
BOAT ALERT!
Latitude
```

After emergency clearance: `Boat Data` , `Cleared`

The LCD therefore serves as a local monitoring interface.

4.2.5 Emergency Detection Logic

The SOS button is connected to GPIO D3 using the internal pull-up resistor.

```
if(lastButtonState==HIGH &&
state==LOW &&
!emergencySent)
{
    triggerEmergencyAlert();
}
```

Once pressed:

- Emergency flag becomes TRUE
- Red LED turns ON
- Green LED turns OFF
- LCD displays Boat Alert
- Current GPS location is uploaded
- Blynk status changes to **BOAT ALERT**
- Buzzer starts sounding

This prevents multiple alerts from being generated until the emergency is cleared.

4.2.6 Alarm System

The buzzer operates intermittently using software timing.

```
void beepBuzzer()
```

The buzzer alternates between:

- 200 ms ON
- 300 ms OFF

This creates a continuous warning tone while minimizing power consumption.

4.2.7 Blynk Cloud Communication

The ESP8266 uploads information to different virtual pins.

Virtual Pin	**Function**
V0	Latitude
V1	Longitude
V2	Date
V3	Status
V4	Emergency Reset
V5	Speed
V6	Google Maps Link
V7	Heading

This allows remote users to monitor the boat through the Blynk mobile application.

4.2.8 Emergency Reset

The emergency can be cleared remotely through the Blynk application.

```
BLYNK_WRITE(VPIN_CLEAR)
```

When activated:

- Alarm stops
- LEDs return to normal
- GPS values reset
- LCD displays "Boat Data Cleared"
- System resumes normal tracking.

4.3 Testing of the Project

The developed system was tested under different operational conditions to verify its functionality.

Figure 4.1 Physical System Testing

4.4 Chapter Summary

This chapter presented the implementation results and performance evaluation of the Smart IoT Solution for Monitoring and Tracking Small Sailing Boats in Mogadishu Coastal Areas. The logical design showed that the ESP8266 software successfully integrates GPS data acquisition, Wi-Fi communication, LCD display, emergency detection, and cloud-based monitoring. The physical design demonstrated the correct interfacing of all hardware components, including the ESP8266 NodeMCU, NEO-6M GPS module, LCD display, SOS button, LEDs, buzzer, and power supply. Functional testing confirmed that the system reliably performed real-time tracking, displayed boat information locally, transmitted location data to the Blynk platform, generated emergency alerts, and allowed remote alert reset. These results indicate that the proposed system is effective, reliable, and suitable for improving the safety and monitoring of small sailing boats in the Mogadishu coastal areas.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the conclusion and recommendations of the study entitled **"A Smart IoT Solution for Monitoring and Tracking Small Sailing Boats in Mogadishu Coastal Areas."** It summarizes the project's contributions, evaluates the achievement of the research objectives, discusses the limitations encountered during the development and testing phases, and proposes possible future enhancements. Finally, recommendations are provided to improve the system and support its practical implementation for enhancing maritime safety along the Mogadishu coastline.

5.2 Contribution

The developed Smart IoT Boat Monitoring and Tracking System contributes significantly to improving the safety and monitoring of small sailing boats operating in Mogadishu coastal areas. The main contributions of this project include:

- Development of a low-cost IoT-based boat monitoring and tracking system suitable for small fishing and sailing boats.
- Integration of the ESP8266 NodeMCU with the NEO-6M GPS module for real-time location tracking.
- Implementation of wireless communication through the Blynk IoT platform, enabling remote monitoring using smartphones.
- Development of an emergency SOS alert system that transmits the boat's location when assistance is required.
- Provision of a local monitoring interface using a 16×2 I2C LCD display for displaying system status and GPS information.
- Integration of visual (LEDs) and audible (buzzer) alarm systems to notify users during emergency situations.
- Generation of Google Maps links to allow rescue teams or family members to locate the boat accurately.

5.3 Achievement of Objectives

The study successfully achieved its research objectives as summarized below.

Objective 1: To design and develop an IoT-based boat monitoring system.

Achievement:

This objective was successfully achieved by designing and implementing a complete hardware system consisting of:

- ESP8266 NodeMCU
- NEO-6M GPS Module
- LCD 16×2 I2C Display
- SOS Push Button
- LEDs
- Buzzer
- Blynk IoT Platform

The hardware components were correctly interfaced and functioned as expected.

Objective 2: To monitor the real-time location of small sailing boats.

Achievement:

The GPS module successfully acquired real-time geographic coordinates (latitude and longitude), which were displayed on the LCD and uploaded to the Blynk cloud every five seconds. The system also generated a Google Maps link that allowed users to locate the boat accurately.

Objective 3: To develop an emergency alert system.

Achievement:

The emergency subsystem successfully detected SOS button presses and automatically:

- Activated the buzzer
- Turned ON the red LED
- Turned OFF the green LED
- Displayed an emergency message on the LCD
- Uploaded the boat's current GPS coordinates to the Blynk cloud

This objective was fully achieved.

Objective 4: To improve maritime safety using IoT technology.

Achievement:

The developed prototype demonstrated that IoT technology can improve maritime safety by providing:

- Continuous monitoring
- Real-time tracking
- Fast emergency notification
- Remote supervision

The system can reduce rescue response time during marine emergencies.

5.4 Limitation

Although the developed system performed successfully, several limitations were identified during the study.

1. Internet Dependency

The system requires a stable Wi-Fi or Internet connection to communicate with the Blynk cloud. In offshore areas with poor network coverage, real-time monitoring may not be possible.

2. GPS Signal Availability

The GPS module requires clear visibility of satellites. During poor weather conditions or when obstructed, GPS accuracy may decrease or location updates may be delayed.

3. Limited Communication Range

The prototype relies on Wi-Fi connectivity. Small boats traveling beyond Wi-Fi coverage cannot continuously transmit location data unless another communication technology (such as GSM, LoRa, or satellite communication) is used.

4. Power Supply Constraints

Continuous operation of the ESP8266, GPS module, LCD, and Wi-Fi consumes electrical power. Long-term deployment requires a reliable power source or rechargeable battery with sufficient capacity.

5. Prototype-Level Design

The developed system was tested as a prototype and has not yet been deployed for long-term operation under actual marine conditions such as saltwater exposure, heavy rain, and extreme temperatures.

5.5 Future Enhancement

Several improvements can be implemented to increase the functionality and reliability of the proposed system.

Integration of GSM/4G/5G Communication

Replacing or supplementing Wi-Fi with GSM or LTE communication would allow the system to operate in wider coastal areas.

Solar Power System

Installing solar panels and rechargeable batteries would improve energy efficiency and enable long-term operation without frequent battery replacement.

Geofencing

Future versions can include geofencing functionality that automatically alerts authorities when boats move beyond designated fishing zones or safe operating boundaries.

Weather Monitoring

The system can integrate environmental sensors to monitor:

- Wind speed
- Wave height
- Water temperature
- Atmospheric pressure

This would provide additional safety information to boat operators.

Mobile Application

A dedicated Android or iOS application could provide a more user-friendly interface than the current Blynk dashboard, offering customized notifications, maps, and historical trip data.

Cloud Database

Future work can integrate cloud databases (such as Firebase or MySQL) to store historical boat routes, emergency events, and performance logs for analysis.

Multiple Boat Management

The system can be expanded to monitor multiple boats simultaneously through a centralized web dashboard for fishing cooperatives or maritime authorities.

5.6 Recommendations

Based on the findings of this research, the following recommendations are proposed:

1. Government maritime authorities should consider adopting affordable IoT-based tracking systems to improve the safety of small fishing and sailing boats operating in Mogadishu coastal waters.
2. Fishing cooperatives should equip their boats with GPS tracking and emergency alert systems to reduce search and rescue response time during accidents.
3. Future researchers should investigate the use of GSM, LoRaWAN, or satellite communication technologies to provide reliable communication in offshore environments.
4. Waterproof enclosures should be used to protect electronic components from seawater, humidity, and harsh weather conditions.
5. Renewable energy sources, such as solar panels, should be incorporated to ensure continuous operation during long fishing trips.
6. Machine learning and artificial intelligence techniques can be explored to predict potential risks, identify abnormal boat movement, and improve emergency response.
7. Large-scale field testing should be conducted using multiple boats under real operating conditions to evaluate the system's reliability, communication performance, and durability over extended periods.

5.7 Conclusion

This study successfully designed and implemented **A Smart IoT Solution for Monitoring and Tracking Small Sailing Boats in Mogadishu Coastal Areas** using the ESP8266 NodeMCU, NEO-6M GPS module, LCD display, Blynk IoT platform, SOS emergency button, LEDs, and buzzer. The developed system demonstrated the ability to monitor boat locations in real time, display GPS information locally, transmit tracking data to a cloud platform, and provide immediate emergency alerts with accurate location information.

Testing results confirmed that the prototype performed reliably in acquiring GPS coordinates, generating Google Maps links, activating emergency alarms, and supporting remote monitoring through the Internet. These features make the system a practical and cost-effective solution for enhancing the safety of small sailing and fishing boats.

Despite limitations such as dependence on Internet connectivity and GPS signal availability, the project demonstrates the potential of IoT technologies to improve maritime safety and emergency response. With enhancements such as GSM or satellite communication, renewable energy integration, geofencing, and dedicated mobile applications, the proposed system can evolve into a robust solution suitable for large-scale deployment in Somalia and other coastal regions. The study therefore concludes that the proposed Smart IoT Boat Monitoring and Tracking System successfully met its objectives and provides a strong foundation for future research and real-world maritime safety applications.

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